F78AA Actuarial and Financial Mathematics A - Sample solutions to December 2016 Exam

1. Let TWRR be the annual time-weighted return. Then

$$\begin{aligned}(1+\text{TWRR})^3 =& \frac{685}{560}\,\frac{612}{685+54}\,\frac{650}{612}\,\frac{798}{650+41}\,\frac{864}{798}\,\frac{986}{864+89}\\ =& 1.39185\\ \Rightarrow \text{TWRR} =& 11.651\%\text{ p.a.}\end{aligned}$$

2. The equation of value is

$$f(i) := 4s_{\overline{20}|} + 100 - 96(1+i)^{20} = 0.$$

As we can't solve it exactly, we use linear interpolation. Trial-and-error gives

$$f(4\%) = 8.764 \qquad \text{and} \qquad f(5\%) = -22.453.$$

Denote by $i^\star$ the solution to $f(i) = 0$. Then

$$i^\star \approx 4\% + 1\%\,\frac{0 - 8.764}{-22.453 - 8.764} = 4.28\%.$$

3. Post-income tax, the investor receives the fraction $(1-d)$ of each rental payment. The equation of value is

$$\begin{aligned}V_0 =& \frac{r_1(1-d)}{1+i} + \frac{r_1(1-d)(1+g)}{(1+i)^2} + \frac{r_1(1-d)(1+g)^2}{(1+i)^3} + \cdots\\ =& \frac{r_1(1-d)}{1+i}\left(1 + \frac{1+g}{1+i} + \left(\frac{1+g}{1+i}\right)^2 + \cdots\right)\\ =& \frac{r_1(1-d)}{1+i}\,\frac{1}{1-\frac{1+g}{1+i}}\\ =& \frac{r_1(1-d)}{i-g}.\end{aligned}$$

4. (a)

$$\begin{aligned}(Ia)_{\overline{n}|} =& v + 2v^2 + 3v^3 + \cdots + nv^n\\ (1+i)\,(Ia)_{\overline{n}|} =& 1 + 2v + 3v^2 + \cdots + nv^{n-1}\\ \Rightarrow (1+i-1)\,(Ia)_{\overline{n}|} =& 1 + v + v^2 + \cdots + v^{n-1} - nv^n\\ \Rightarrow i\,(Ia)_{\overline{n}|} =& \ddot{a}_{\overline{n}|} - nv^n\\ \Rightarrow (Ia)_{\overline{n}|} =& \frac{\ddot{a}_{\overline{n}|} - nv^n}{i}.\end{aligned}$$

(b) The P.V. at the start of the payment stream is

$$\begin{aligned}\text{P.V.} =& 10v + 12v^2 + 14v^3 + \cdots + 48v^{20} \\ =& 10v + (10+2)v^2 + (10+4)v^3 + \cdots + (10+38)v^{20} \\ =& 10a_{\overline{20}|} + 2v\left(v + 2v^2 + 3v^3 + \cdots + 19v^{19}\right) \\ =& 10a_{\overline{20}|} + 2v(Ia)_{\overline{19}|}.\end{aligned}$$

Calculating at $i = 0.03$, we find

$$a_{\overline{20}|} = 14.87747, \quad \ddot{a}_{\overline{19}|} = 14.75351$$

and

$$(Ia)_{\overline{19}|} = \frac{\ddot{a}_{\overline{19}|} - 19v^{19}}{0.03} = \frac{14.75351 - 19v^{19}}{0.03} = 130.60252.$$

Therefore,

$$\text{P.V.} = 10 \times 14.87747 + 2v \times 130.60252 = 402.37183.$$

Alternatively, P.V. $= 8a_{\overline{20}|} + 2(Ia)_{\overline{20}|}$. As

$$(Ia)_{\overline{20}|} = \frac{\ddot{a}_{\overline{20}|} - 20v^{20}}{0.03} = \frac{15.32380 - 20v^{20}}{0.03} = 141.67616,$$

we get P.V. $= 8 \times 14.87747 + 2 \times 141.67616 = 402.37208$.

5. (a) Let P be the monthly repayment. The monthly interest rate is $j := 1.05^{1/12} - 1 = 0.003274$ and there are 300 months in 25 years. The equation of value is

$$P = \frac{100\,000}{a_{\overline{300}|}^{0.003274}} = \frac{100\,000}{190.87122} = \text{£}523.91.$$

(b) As the interest paid in the first month is

$$100\,000 \times 0.003274 = 327.40,$$

the capital repaid is

$$523.91 - 327.40 = \text{£}196.51.$$

(c) The capital outstanding after 20 years (with 60 months remaining) of repayments is

$$Pa_{\overline{60}|}^{0.003274} = 523.91 \times 54.39392 = \text{£}28\,497.52.$$

The total amount of repayments (no discounting) payable in the next 60 months is

$$60P = 60 \times 523.91 = \text{£}31\,434.60.$$

Therefore, the total amount of interest paid is

$$31\,434.60 - 28\,497.52 = \text{£}2\,937.08.$$

(d) The capital outstanding just before the last repayment is

$$Pa_{\overline{1}|}^{0.003274} = 523.91/1.003274 = \text{£}522.20.$$

The interest included is

$$522.20 \times 0.003274 = \text{£}1.71.$$

(e) If repayments are made less frequently than monthly, the total annual repayment increases since *interest* payments are made later.

6. (a) The net interest rate for the payments from the electricity company is $1.03/1.04 - 1 = -0.00962$.
The NPV at the start of the project is

$$\begin{aligned}
\text{NPV} &= -8\,000 + 300v^2 + 300 \times 1.04v^3 + 300 \times 1.04^2v^4 + \cdots + 300 \times 1.04^{19}v^{21} \\
&\quad + 150v + 150v^2 + 150v^3 + \cdots + 150v^{20} \\
&= -8\,000 + 300v^2\left(1 + 1.04v + 1.04^2v^2 + \cdots + 1.04^{19}v^{19}\right) \\
&\quad + 150\left(v + v^2 + \cdots + v^{20}\right) \\
&= -8\,000 + 300(1.03)^{-2}\ddot{a}_{\overline{20}|}^{-0.962\%} + 150a_{\overline{20}|}^{3\%}.
\end{aligned}$$

As

$$a_{\overline{20}|}^{3\%} = \frac{1 - 1.03^{-20}}{0.03} = 14.87747$$

and

$$\ddot{a}_{\overline{20}|}^{-0.962\%} = \frac{1 - 0.99038^{-20}}{1 - 0.99038^{-1}} = 21.95669,$$

we find

$$\text{NPV} = -8\,000 + 300(1.03)^{-2} \times 21.95669 + 150 \times 14.87747 = 440.51.$$

(b) Since the outgo precedes the income and the project has a positive NPV, the IRR must be greater than 3% to bring the NPV down to zero.

7. (a) Work in £000s and use half-years, so that the half-yearly effective interest rate $j := 1.06^{1/2} - 1 = 0.02956$. Let t be the first time (in half-years) that the discounted income exceeds the discounted loan value, i.e.

$$5a_{\overline{t}|}^{2.956\%} \geq 120.$$

Rearranging,

$$5\frac{1 - (1+j)^{-t}}{j} \geq 120 \qquad \Rightarrow \qquad t \geq \frac{\ln\left(1 - \frac{120}{5}j\right)}{\ln\left((1+j)^{-1}\right)} = 42.426 \text{ half-years.}$$

In years, the minimum time is 21.213 years. So, as the annuity payments are half-yearly, the DPP is 21.5 years (or 43 half-years).

(b) Use half-years. The profit remaining after the loan has been repaid at the DPP time is

$$\begin{aligned} 5\,000 s_{\overline{43}|}^{2.956\%} - 120\,000 \times (1.02956)^{43} &= 5\,000 \times 84.56104 - 419\,954.92 \\ &= 422\,805.20 - 419\,954.92 \\ &= 2\,850.28. \end{aligned}$$

so the accumulated profit after 30 years is (now using the annual 4% effective interest rate as the loan has been repaid, which is $1.04^{1/2} - 1 = 0.01980$ per half-year)

$$\begin{aligned} 2\,850.28 \times (1.01980)^{17} + 5\,000 s_{\overline{17}|}^{1.980\%} &= 3\,977.80 + 5\,000 \times 19.97885 \\ &= 3\,977.80 + 99\,894.25 \\ &= 103\,872.05. \end{aligned}$$